

Haomin LUO

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EDUCATION

University of Cambridge, St John's College

Oct 2024- Jul 2029

PhD: Information Engineering

United Kingdom

Research Group: Energy Efficient Cities initiative & The Alan Turing Institute | Supervisors: Prof. Ioannis Lestas & Ruchi Choudhary

Research Focus: AI & Energy Finance

1. Multi-agent deep reinforcement learning (RL), researching agent communication mechanisms, robustness and explainability
2. Applied RL algorithm as a model-free, multi-objective control method on time-series data for demand response scheduling of microgrids, VPPs, focusing on its applications in energy finance, especially quantitative trading in electricity markets

The Hong Kong University of Science and Technology

Sep 2022- Jun 2024

MSc: Intelligent Building Technology and Management

Hong Kong

GPA: 3.96 (Top 5%) | Excellent Student Scholarship | 2nd place in NeurIPS Challenge 2023 | Supervisor: Prof. Zhe (Walter) WANG

Southwest Jiaotong University

Sep 2016 – Jun 2020

BEng: Civil Engineering

China

EMPLOYMENT

Huatai Securities

Oct 2022 – Feb 2023

Equity Research Intern, TMT Group

Hong Kong

- Industry Tracking: Monitored US and HK tech sectors, focusing on semiconductors, display panels, and consumer electronics. Tracked major EV players including Tesla, BYD, and the “Wei-Xiao-Li” trio (NIO, XPeng, Li Auto)
- In-Depth Research: Initiated coverage on Lianchuang Electronic (002036.SZ) with a comprehensive investment report; co-authored an English report for BOE Varitronix (0710.HK); produced **5 thematic reports** covering smart cockpits (LiDAR, chips) and monthly trends in autonomous driving and power batteries

Soochow Securities

Apr 2022 – Aug 2022

Equity Research Intern, Real Estate Group (New Fortune Top-Ranked Team)

China

- Tracked central & local real estate easing policies amid nationwide credit risks, monitored cement, glass and other building materials' prices & capacity, and **analyzed 315 listed firms'** competition, supply-demand, inventory and turnover to build investment logic.
- Updated upstream & downstream industry data via Wind, covering local government new bonds, new contract values, PV glass and soda ash energy spreads; built earnings & valuation models for core enterprises and **released over 20 weekly/monthly analysis reports**
- Collected land acquisition firms' financial metrics for land price sensitivity research, analyzed inventory and destocking cycles, and calculated 5-year comprehensive financing costs for **114 real estate enterprises** including Poly Development (600048.SH). Expanded team research coverage and independently finished a 10,000-word in-depth report on integrated-stove leader Yitian Smart (300911.SZ).

Meritco Services

Jan 2022 – Mar 2022

Part-Time Assistant (PTA), Hard Tech Group

China

- Assisted project managers in serving PE/VC and other investment institutions with investment consulting on frontier tech sectors such as EV batteries, AI image-processing chips, and cloud-computing data centers; designed quantitative information-source methodologies based on preliminary research and **customized 10+ expert interview** outlines combined with desk research.
- Structured **80+ hours of interview transcripts**, and after data cleaning and verification, applied analytical frameworks such as Porter's Five Forces and SWOT to produce **5 investment reports**.

AECOM Consulting

Jan 2021 – Jan 2022

Assistant Engineer (Full-time)

China

- In the **Global Delivery Center (GDC)**, responsible for the intelligent design, operation and maintenance of infrastructure assets around the world, providing sustainability consulting services, including full-lifecycle energy consumption and carbon footprint analysis (ESG)
- Based on 3D information data, applied Big Data, BIM, and Cloud Computing techs to enhance the level of information management on cloud-based visualization platforms; awarded **MVP for the Sydney Gateway project** in Australia, and participated City Rail Link in New Zealand, and Mandai Safari Park in Singapore

SKILLS

- IELTS 8.0 | CFA Level I Candidate | Wind, iFind, Bloomberg | Python (numpy,etc) | Gemini, Codex, Cursor | C1 Driving License
- Fashion Model (Agency: Lacocomodels) | Executive Member of the Entrepreneurship Club | LMBC Rower | Sailing | Violinist

RESEARCH

From Centralization to Intelligent Cooperation: Integrating Adaptive Grouping into MADDPG for Distributed Energy Storage Coordination in Smart Microgrids

Oct 2024 – Jun 2026

Based on First Year report, will be put on arXiv for Publication Later

- This work tackles the scalability-coordination dilemma in distributed energy resource management for smart microgrids, and proposes HTDE-MADDPG — a novel MARL framework that integrates an adaptive grouping module into MADDPG.
- It adopts Graph Attention Networks for information aggregation and Gumbel-Sigmoid sampling for differentiable dynamic cluster formation, achieving strong inter-agent collaboration while retaining decentralized execution.
- Validated on real-world microgrid data in the CityLearn environment, the **algorithm reduces operational cost by 46.6%**, peak demand by 37.0%, carbon emissions by 38.7% and load ramping by 22.3% against the baseline, consistently outperforming DTDE-MADDPG, CTDE-MADDPG and MPC benchmarks with faster convergence and lower communication overhead.

Artificial Intelligence in Virtue Power Plants(VPPs) and Microgrids Controls

Sep 2022 – Jun 2024

Project Leader in GREAT Smart Cities Institute (GSCI) ., HKUST., Supervised by Prof. Zhe (Walter) WANG

- This project aims to develop advanced controllers (e.g., model predictive control and reinforcement learning) to coordinate the load consumption of different households and to operate the thermal energy storage system to minimize the utility costs given the time-of-use utility structure and to minimize the CO2 emissions given varying carbon intensity of the grid
- The **NeurIPS 2022: CityLearn Challenge's** 17 buildings' one year operation data from the Sierra Crest home development in California were used as a virtual testbed and dataset, including PV generation, Weather data, Electricity demand, etc
- Trained multiple advanced deep reinforcement learning control algorithms (DDPG, SAC, PPO, TD3) to manage the battery charging/discharging behavior, made the final result performs 13% better than the baseline
- Designed several corresponding KPIs (e.g. Average daily peak, Ramping, Zero net energy, Carbon emissionsd, etc.,) for the comparative study between different algorithms at the level of single building-level and grid-level, respectively, and it was found that the TD3 algorithm had the best overall performance, and was ahead of the DDPG by 3%-27% in proposed KPIs

CFD Simulation of Wind Tunnel Tests for High-rise Buildings

Apr 2018 – Jun 2019

Research Assistant in XNJD-3 Lab (world's largest atmospheric boundary layer wind tunnel).. Supervised by Prof. Shixiong Zheng

- Participated in the CFD simulation of the wind resistance tests of high-rise buildings & large-span bridges under different wind directions & speeds, and the wind-induced catastrophe research of urban infrastructures in wind tunnel
- Responsible for the finite element analysis of buildings' aeroelastic models under specific scale ratios and the simulation of the turbulent flow field in the aerodynamic experiment, to solve wind vibration problems such as flutter, galloping and buffeting.
- Successfully completed the wind tunnel test of the **1915 Turkish Çanakkale Bridge (world's longest bridge)** within four months; Other projects including Shanghai Lujiazui Financial Trade Zone skyscraper cluster and Hålogaland Bridge in Norway

Green Building Design and Energy Simulation for Airport Terminals

Jun 2017 – Sep 2017

Research Assistant in China State Construction Engineering Corporation (CSCEC) of the Chengdu International Airport project

- Committed to enhancing the environmental sustainability of airport terminals while simultaneously reducing their operational costs. throughout the entirety of the project's design and construction process, responsible for examining the overall energy consumption and carbon footprint of the T1 terminal (1.26 km²) and provide renewable energy feasibility solutions
- Conducting microclimate analysis of the site and used EnergyPlus to analyze energy consumption; through glare analysis of building facades, indoor thermal comfort analysis, and indoor lighting environment analysis, we used Ecotect, Fuzor and Sketchup to simulate the building's layout, ventilation, light intensity and calculated its best orientation
- The project won the 5th International BIM award (hosted by BSHK) and the excellence award of **the 2020 Bentley Competition**

PUBLICATION & AWARDS

- Haomin, LUO., A UVC Purifier for Improving IAQ in Building Environment. Patent Application Number: ZL 2013 3 0320199.2
- Haomin, LUO., Indoor Air Purification Device. Patent Application Number: ZL 2013 2 0094064.3 (Authorized)
- Haomin, L., Choudhary, R., 2026. "Integrating Adaptive Grouping into MADDPG for Distributed Energy Storage Coordination in Smart Microgrids". soon on arXiv)
- 03/2024 MSc Excellent Student Scholarship (Top 5%);
- 11/2023 2nd Place in the 2023 NeurIPS CityLearn Challenge (Top 2);
- 05/2021 Most Valued Player in Sydney Gateway Project of AECOM (Top 1);
- 12/2018 Mingcheng Scholarship of Southwest Jiaotong University (Top 10%);
- 06/2018 2017-2018 Outstanding Individual of Southwest Jiaotong University (Top 5%);
- 04/2018 6th Place in 2018 ASCE Mid-Pacific Student Conference Concrete Canoe Competition (International Award);
- 03/2017 4th Place in the 2017 ASCE Southeast Student Conference Steel Bridge Competition (International Award)